BT HealthNet — an early intranet case study

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Like the Internet, the intranet is much hyped. BT recognised the emerging importance of the Internet and its technology and commenced investment in its own intranet some 5 years previous. BT's experience in both building intranets and the Internet service provision business has led to the development of quite possibly the largest community-of-interest Intranet in Europe — the UK's NHS-wide network or NHSnet. BT HealthNet represents the core of NHSnet and this paper illustrates how its components and services provide a first-class platform for the delivery of healthcare information and applications.

1. Introduction

BT's HealthNet is the NHS-wide network standard for data communications. It is a managed Internet protocol (IP) [1] service, incorporating facilities to securely interconnect all NHS sites and thus distribute existing and future applications within sites or across the NHS as a whole, integrating local area networks (LANs) and replacing existing private wide-area networks.

In the past a number of different connections were required to enable NHS organisations to communicate with local healthcare partners, or natural community. A single connection to NHS *net*, through HealthNet [2], provides for part or all of an NHS organisation's network infrastructure; it also offers the added bonus of providing a secure gateway to the Internet.

BT HealthNet WebWorld [2] is a web-hosting service that offers NHS and non-NHS organisations World Wide Web publishing on NHS *net* (see Fig 1), at competitive rates to those available on the public Internet but with improved access speeds, functionality and the security of a private network. By mounting Web-based information and applications in a secure, networked environment, customers are free to concentrate on supporting their own business objectives, leaving the supplier to take on complex technical and operational activities on their behalf.

This paper sets out the technical platforms that make-up HealthNet services and illustrates how reuse of off-the-shelf Internet technology can be used to build a significant community-of-interest network that will ultimately be used to reduce costs and increase efficiencies of the delivery of healthcare in the UK.

2. Background

BT's investment in HealthNet is in response to the NHS executive's commitment to reducing costs and increasing efficiencies within the National Health Service [3, 4]. An important additional component to HealthNet is the NHS message handling service, provided by BT Syntegra. This X.400 service is the first step in automating business processes within the health service, specifically in the administration of the delivery of primary healthcare, between GP (general practitioner) fundholder, health authority and healthcare provider. The message handling service is another BT product in its own right and is not covered in this paper.

HealthNet WebWorld is a complementary product to HealthNet whose primary objective is to stimulate interest in the network — through its third-party Web content — and to facilitate the delivery of information and applications.

3. HealthNet

3.1 Architecture

The design of HealthNet has been driven by customer requirements for a National Health Service private cor porate network [5]. Access to the network is provided via switched (PSTN and ISDN) and leased lined methods; switched access, by its nature had to be accompanied by strong authentication access control mechanisms [6, 7]. Key quality of service requirements included resiliency, proactive network management and managed router options. In recognition of the importance of interfaces to the outside world, a secure gateway to the Internet was also required.

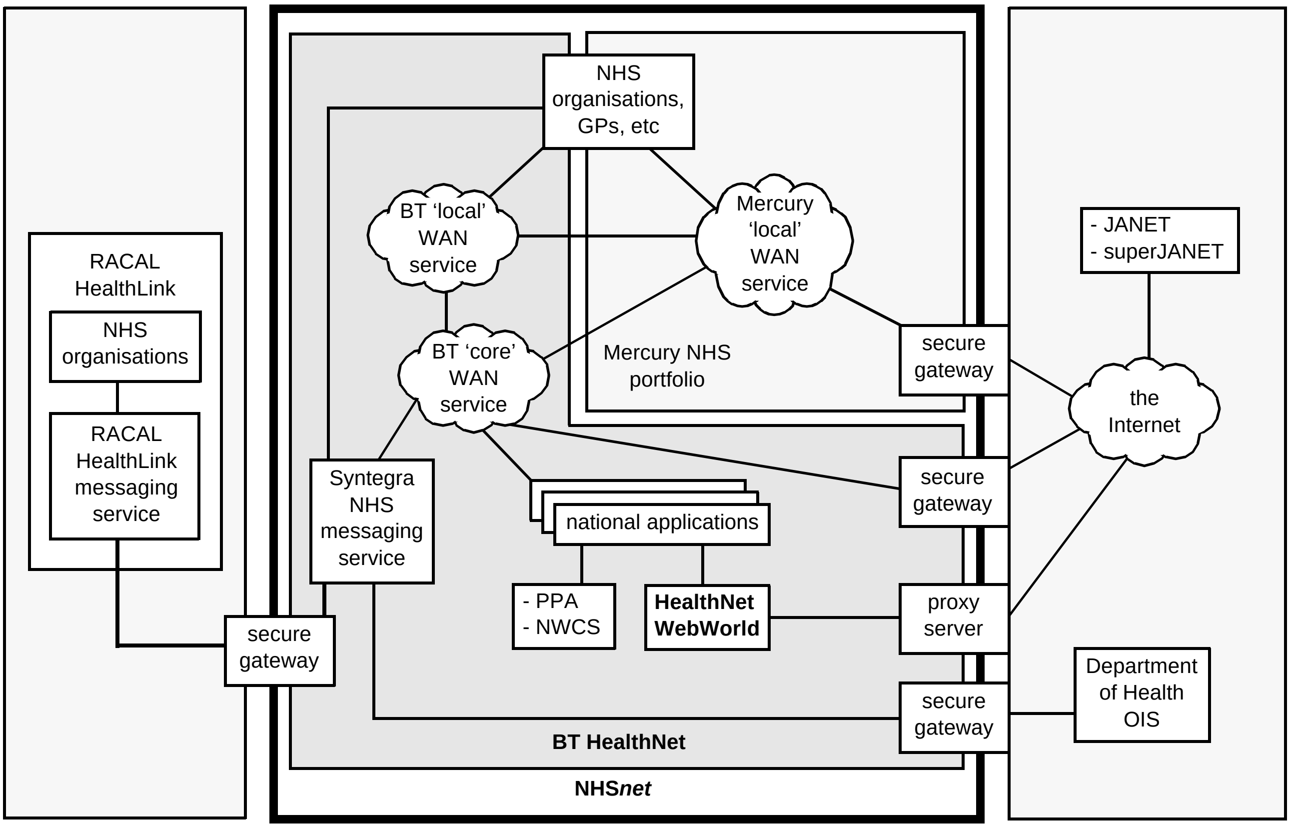

Fig 1 NHS*net* and its components.

The BT HealthNet IP service is based upon a dedicated access infrastructure provided at 11 points of presence (PoPs) throughout the UK (see Fig 2). All PoPs are inter-connected by means of a core network based upon the BT switched multi-megabit data service (SMDS). This means that each PoP is only one network hop from every other PoP.

This network architecture ensures that additional PoPs and growth within the PoP can be achieved as use of the network increases. Customers of the network have differing requirements dependent on their end systems and applications, and to allow customer solutions to be optimised, a number of service options are offered. Connection methods range from 2.4 kbit/s PSTN to 2 Mbit/s dedicated access, with the customer able to use a stand-alone PC, their own router, or a BT managed router.

3.2 Network implementation

Customer access to HealthNet

There are six types of HealthNet access:

- dedicated (leased line) access,
- resilient (dual leased line) access (using loadsharing),
- dedicated (leased line) access with ISDN back-up,
- ISDN dial-up access using router,
- PSTN/ISDN dial-up access using PC/workstation — static IP address,
- PSTN/ISDN dial-up access using PC/workstation — dynamic IP address.

Broadly, these can be categorised as dial-in or dedicated methods whose security classifications can be found in Table 1.

Table 1 Security classification of access connection types.

Type	Description	Security classification
ISDN (multiple user)	The site is using PPP CHAP to the PoP and PPP CLI from PoP.	Trusted
(ISDN/PSTN) Strong authentication access	The user is using a smart card to gain access to the service.	Trusted
(ISDN/PSTN) Standard authentication access	The site is using PPP to gain access to the service.	Non-trusted
Dedicated access	The site is using a dedicated leased line to access the service, and the NHS has identified the site as trusted	Trusted or Non-trusted as specified
Non-NHS entities	Non-NHS entities are organisa-tions that are attached to the NHS network, but are not part of the NHS.	Non-NHS

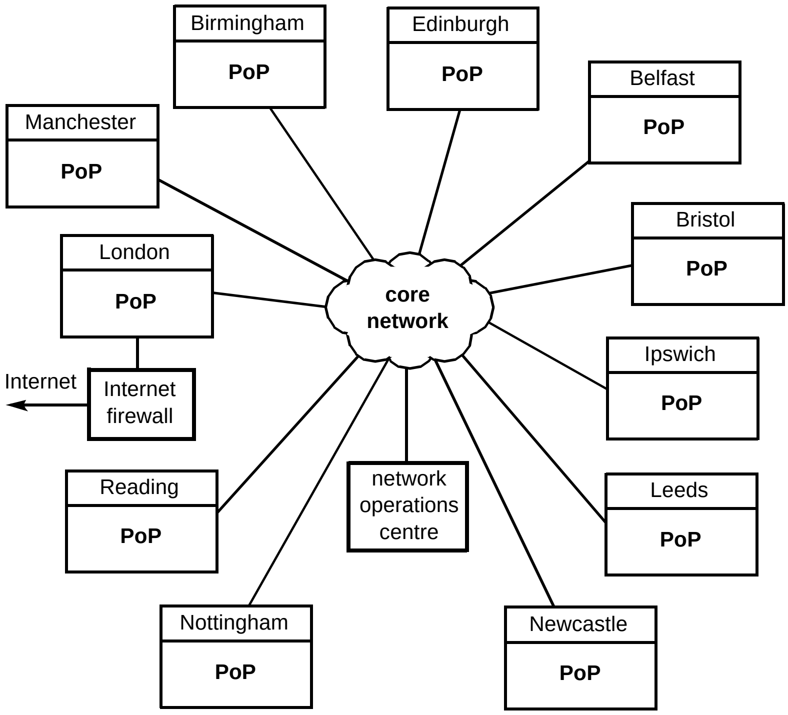

Fig 2 Core HealthNet network.

Dial access

Dial access is available via PSTN at 2.4 kbit/s to 28.8 kbit/s or ISDN at 64 kbit/s (see Fig 3) .

Dedicated access

Dedicated access to the HealthNet IP service is provided at speeds of 64 kbit/s, 128 kbit/s, 256 kbit/s, 512 kbit/s, 1.024 Mbit/s, and 2.048 Mbit/s. Where a site requires a higher level of access, resilience options are available with 64 kbit/s ISDN back-up and dual resilient (see Fig 4 for example).

All access methods, with the exception of PSTN, are offered with a BT managed router or as an interface to which a customer-owned router can be connected, e.g. see Fig 5.

Customer-addressing scheme

NHS organisations requiring multiple hosts are allocated a classful address (i.e. the minimum of a full Class C). Organisations requiring a small number of hosts (less than 30) are allocated addresses from a reserved block as the IP address space can be sub-netted to improve efficiency of address allocation.

PSTN/ISDN dial-up users are allocated static or dynamic IP addresses. A static IP address is required for strong authentication access and will necessitate the user dialling into the same PoP at all times to ensure that the stringent NHS security policies are upheld. For back-up ISDN connections, sites with a pre-allocated NHS IP address use a full class C address.

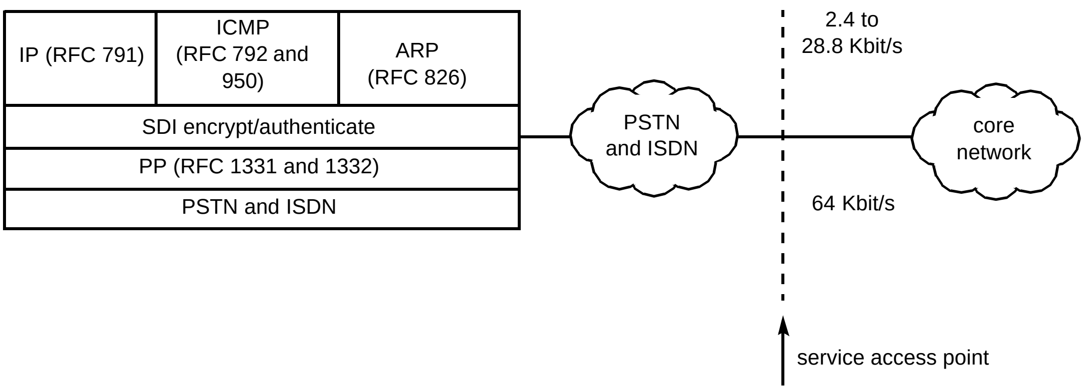

Fig 3 PSTN and ISDN access to HealthNet.

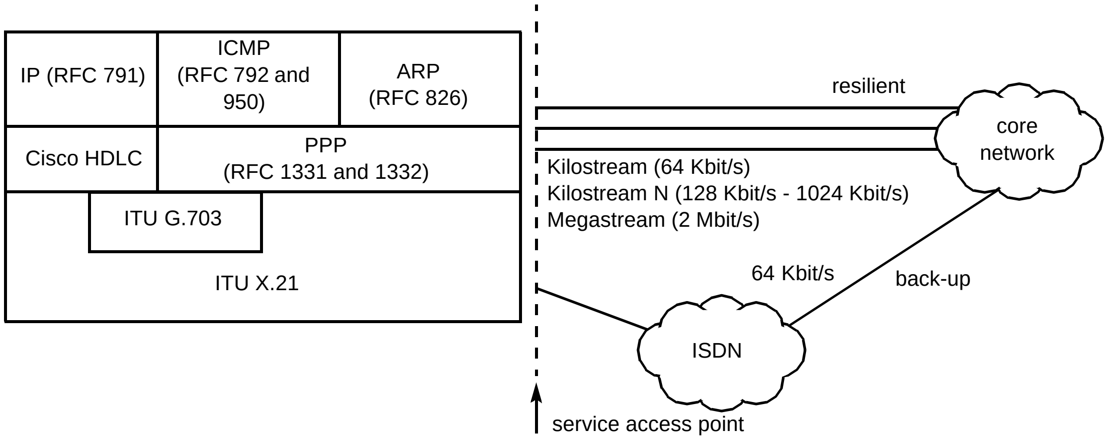

Fig 4 Resilience and back-up dedicated access.

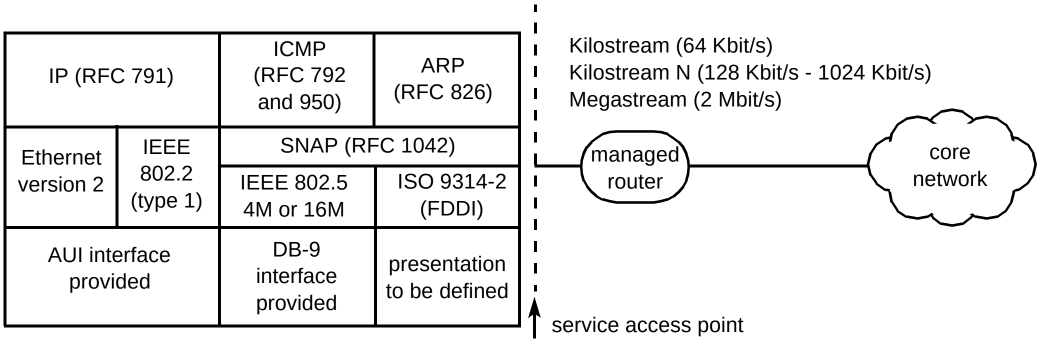

Fig 5 Dedicated access to HealthNet with managed router.

Security mechanisms

As you would expect from a private network, there are multiple security mechanisms to cope with every eventuality, including the following.

- TCP/IP filtering

  Packet filtering is used to control data transfer by monitoring, filtering source and destination addresses to protect against masquerade, and route filtering to control route address distribution for protection against theft of service.

- The application level firewall

  This prevents any access to HealthNet from outside sources, making the network invisible to Internet users outside HealthNet.

- Standard authentication

  This protects access to HealthNet at the most basic level, consisting of simple password verification of the user's identity before access to the network is granted. This method of authentication only provides access to the NHS messaging service.

- Strong authentication (access control)

  This involves the end-user having a proprietary smart card product. The card has a processor into which a 'seed' (a pseudo-random number) is programmed. This is then encrypted using the time since the 'birth' of the card as a key. The password (a 6-digit number) is recalculated every 60 seconds. This is validated along with a 4-character PIN known only to the user to complete the authentication process at the access control server.

- Calling line identifier (CLI)

  Where access to the network is via a router and ISDN, calls are stamped with a CLI which is unique to the ISDN line setting up the call.

- Challenge handshake authentication protocol (CHAP)

  This is a security feature available on devices supporting the point-to-point protocol (PPP) which prevents access from unauthorised devices.

For access to the network, the five types of user access are identified in Table 1 for security purposes, each type being assigned a security classification.

Trusted and **non-trusted** apply on a per-connection basis and as such all users on that connection have the same security classification. The use of ISDN dial-up between routers using a combination of CHAP and CLI is considered to be as secure as a fixed link between routers and is classified as trusted. A single PC connected to the network via standard authenticated dial-in is classed as non-trusted and a single PC connected to the network via strong authentication dial-in is classed as trusted.

HealthNet has its own dedicated CHAP authentication server and strong authentication access control server (which is used as a back-up domain name server (DNS)). The DNS is configured to run the CHAP and access control server processes in order to provide a back-up in case the main server fails.

The key to the security of any LAN protocol network is in the efficient management of the access lists for the routers which control the flow of traffic through the network from source to destination. HealthNet offers an efficient and proven method of access management for LAN protocols, already widely in use where security is crucial. Internal TCP/IP filtering is configured within the receiving routers which vet the stated source address of each incoming IP packet. This ensures that the stated origin address is consistent with that access point, ensuring a sound basis for the traffic control provided by the source address filters. For destination address filtering the network routers provide the second phase of traffic filtering. These are offered in three categories:

- messaging only — where a user has not met the criteria laid down by the NHS for full network membership, they have restricted access to the network facilities of the NHS messaging service only,
- full service access — where a user has fully met the criteria laid down by the NHS for full network membership, they have complete access to all the network facilities,
- PSTN/ISDN dial-up — where a user has fully met the criteria laid down by the NHS for PSTN/ISDN dial access to the network, they have complete access to all the network facilities.

The classification of a customer's access to the network is subject to their status as an NHS or non-NHS organisation and as such is governed by the NHS Executive code-of-connection [8, 9]. In addition, there is a security transition agreement and a third party access agreement.

PoP design

Figure 6 shows an example of how each PoP is constructed. While all the PoPs are designed to be scalable, the differences between PoPs of the expansion capabilities are not shown.

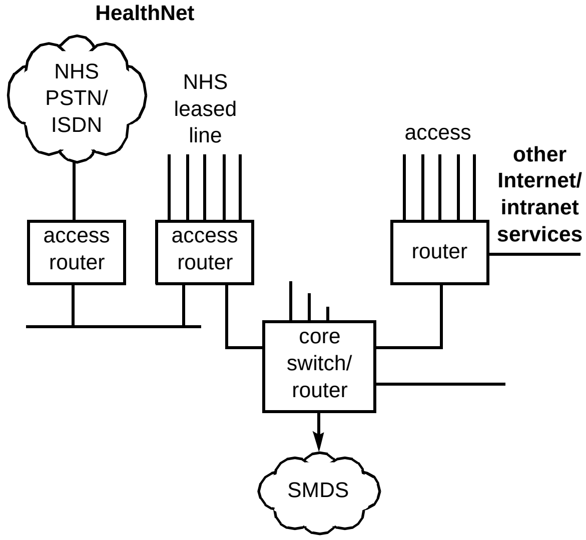

Fig 6 Example PoP design

Routeing/resilience design

The routeing strategy for HealthNet is enhanced interior gateway routeing protocol (EIGRP) between the PoPs (i.e. over SMDS) with RIP over the HealthNet Ethernet segment between the access router and core router at the PoP. Static routes are configured for each customer access.

Routeing between HealthNet and the customer site is direction dependent. For access to an NHS end-site, static routes are used for each access circuit and no routeing protocols are used between the PoP and the end-site. For access to the HealthNet PoP, end-sites use default routes towards HealthNet and only registered addresses are permitted access.

A back-up mechanism (back-up router and ISDN path) is provided in case of failure of the SMDS link or core router. The back-up mechanism has a major influence on the routeing strategy. During normal operation, routeing information is advertised via direct SMDS links as well as via a protocol 'tunnel' to a neighbouring PoP. This ensures that if the back-up ISDN link becomes activated the routeing strategy learnt by the back-up routers becomes operational from the point of failure. If the core router at a PoP fails, an ISDN connection is set up between the back-up routers at each PoP. CHAP is used to authenticate the connection, and once operational, all routeing information and data traffic is passed directly over the ISDN link between partner PoPs.

Each PoP has a partner PoP. This creates a circular topology as illustrated in Fig 7 which shows how routeing information and data traffic is transferred after an equipment failure at one of the PoPs.

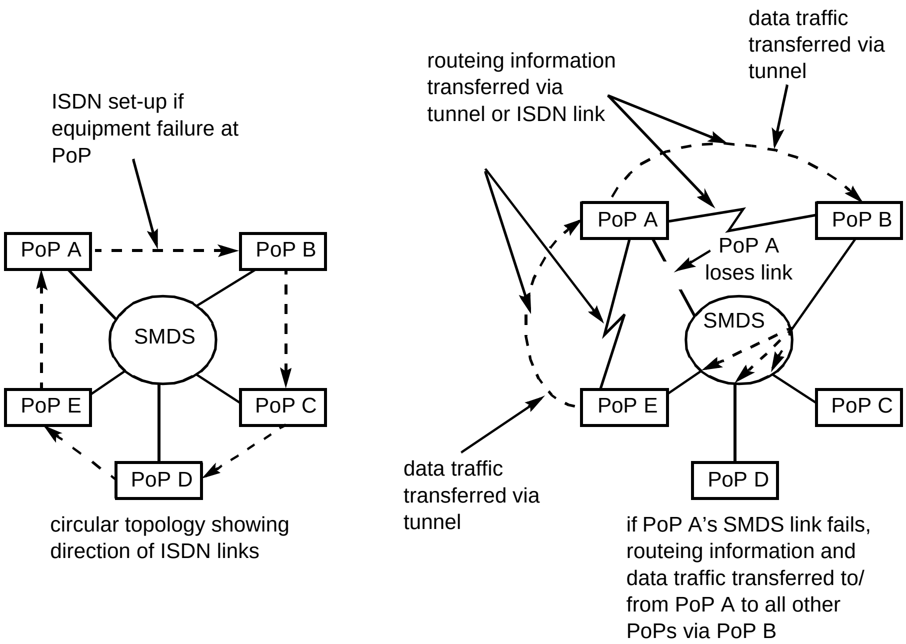

Fig 7 Circular topology for back-up mechanism.

Domain name service

BT operates and administers the server that provides the 'nhs.uk' domain (www.nic.uk/nhs.uk.html) for NHS *net* on behalf of the NHS and, as such, provides the primary DNS nameserver for the whole domain and has delegated authority for the IN_ADDR.ARPA domains corresponding to each block of NHS addresses from RIPE.

The services provided by the DNS include DNS queries for local DNS servers or hosts, and the provision of a secondary nameserver in the cases where larger NHS organisations establish their own primary DNS server. Some organisations may, however, provide their own secondary nameservers.

The HealthNet DNS is responsible for the top levels in the nhs.uk hierarchy (down to a minimum of sub-division level, which has the format 'sub-division.nhs.uk', e.g. 'nthames.nhs.uk'). NHS organisations have responsibility for primary nameserver duties at the lower levels for the 'nhs.uk' domain. However, the HealthNet nameserver has pointers to all other primary nameservers operated by NHS organisations and periodically obtains copies of all the host entries (as part of its secondary nameserver duties). Consequently, BT's nameserver has a complete copy of all DNS entries in the 'nhs.uk' hierarchy that are resident on NHS*net*.

The HealthNet external DNS is aware of route nameservers and popular UK nameservers on the Internet. Thus all DNS queries to external sites are resolved on behalf of the internal DNS via the external DNS querying Internet nameservers for the appropriate name/address mapping (see Fig 8). The only DNS entries visible outside the NHS*net* are contained in the external DNS for NHS on the firewall gateway [10].

The types of entry stored in the external DNS are:

- name/address mappings of accessible NHS domains as identified by the NHS,
- the 'nhs.uk' domain with IP address of the firewall gateway.

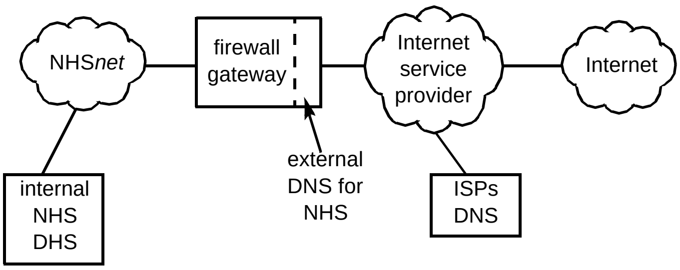

Fig 8 Internal and external DNS servers for NHS.

Internet Access

Restricted access to the Internet is available to customers of HealthNet, this is achieved by means of an application-level firewall gateway. The firewall gateway uses a network address-translation feature. This means that the firewall's external IP address is the only address that is visible to the Internet, as the firewall gateway re-maps all outbound connections so that the Internet sees all packets as originated from the firewall itself. Thus all internal NHS addresses are mapped to a single IP address, namely the IP address assigned to the high-speed serial port on the firewall gateway attached to the Internet service provider's router. The firewall performs address conversions so that applications that require acknowledgements or other returns can have these returns indexed by the firewall, thus preventing any attacks that spoof acknowledgements. Figure 9 illustrates the protocol model for the firewall.

Thus the services available to HealthNet users are summarised in Table 2.

Table 2 Internet services available to HealthNet users.

Protocol	Incoming (to HealthNet)	Outgoing* (from HealthNet)
Telnet	Not permitted	No restriction
FTP	Not permitted	No restriction
WWW	Not permitted	No restriction
NNTP	Permitted with restrictions	Permitted with restrictions
SMTP	Permitted (default to NHS messaging service)	Permitted with restrictions
Gopher	Not permitted	Permitted
Finger	Not permitted	Permitted
PING	Not permitted	Permitted

* Acknowledgements from these protocols are subject to address translations.

Network management

HealthNet is managed from a central network operations centre (NOC). The NOC manages the core network infrastructure of switches, routers, primary DNS, user authentication servers and Internet firewall on behalf of the NHS. The services provided by the NOC include management and configuration of core network, addressing and routeing, administration of dial access accounts, customer router configuration, core network capacity monitoring and planning, management of routers provided by BT, and monitoring of detected network access violations.

4. HealthNet WebWorld

4.1 Introduction

HealthNet WebWorld is an instance of the BT WebWorld [11] Web-hosting platform but specifically engineered for the HealthNet network, taking into consideration the particular security requirements that surround the private intranet. The service is aimed at those organisations that want a hosted presence on NHS *net* and those wanting a hosted presence on both the NHS *net* and the public Internet.

The market for HealthNet WebWorld information and applications is divided into professional sectors. Each sector has distinct information needs which drive improved efficiency and quality of patient care. BT HealthNet WebWorld allows content providers to present relevant information in a common format (i.e. HTML) to their target audience, enabling a rapid, efficient and cost-effective interchange of ideas among various groups in the health service. It also encourages those who are not accustomed to technology to integrate it into their daily lives, enabled by the common interface — the Web browser.

Important customers of HealthNet WebWorld are non-NHS organisations (such as academia, pharmaceuticals and suppliers to the NHS) who would not normally be allowed to take full connection to the NHS *net* [9] or may not want to. These organisations now have the ability to up-load their content via a secure route to the server — this is given using secure access (CHAP) via ISDN or PSTN. This class of content provider can only see their FTP directory and their Web site on the server and they have no onward access into HealthNet.

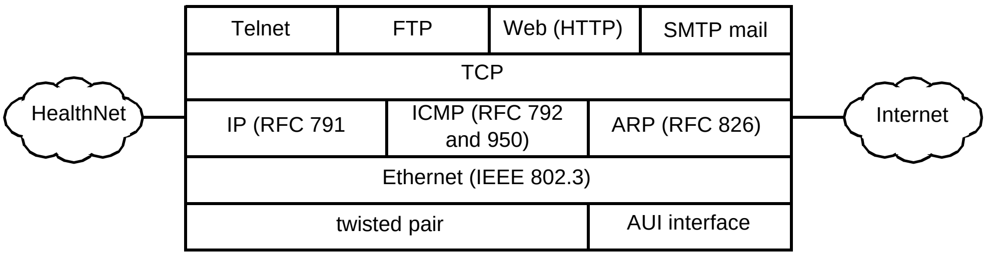

Fig 9 The firewall protocol model.

The advantage to organisations in having a Web site hosted on the NHS*net* only is that they can be sure that only approved users will be able to access the information and that the speed of access which those users perceive will be significantly enhanced over that for the public Internet. However, many organisations will want both intranet and Internet hosting as, like BT itself, they will have some information types (e.g. staff briefings) that they only wish to make available over the intranet [12] but others (e.g. Trust Annual Reports) that they wish to make available to both audiences [13].

4.2 Service features

The Web-hosting platform has much in common with the BT WebWorld platform; in summary, these service levels and features are as follows:

- standard level provides the facility to host data files and make them available via a webserver, server storage space being allocated in 5 MB blocks up to a maximum of 100 MB,

- enhanced level provides the same facilities as standard but is tailored to a larger volume of server space, and additionally provides the ability to set up closed user groups and has many other features available as applications (CGI scripts) that can be driven via Web pages, with server space being allocated in 20 MB blocks up to a maximum of 1 GB,

- custom level offers a bespoke service which includes the hosting of applications for database processing, and includes everything offered by enhanced plus the ability to host extracts from a customer's database that can be driven from a customer's own CGIs (subject to validation and testing).

4.3 Architecture and implementation

The physical platform is located at a secure computer centre location, operated as part of a larger 7 × 24 operation. The heart of the platform is a pair of SUN Sparc 1000s that run the main service. These are protected by hubs, firewalls and routers from four different access mechanisms:

- NHS*net* end users can browse content on the Web platform, and customers who have NHS*net* access can upload content and browse content via this route — there are security policies that influence the design in this respect, in particular, in order to ensure NHS*net* integrity there is no mechanism for routeing between NHS*net* and the other networks connected to the servers,

- Internet users are served by a proxy server that hides the implementation and network details of the HealthNet WebWorld machines from the Internet,

- ISDN or PSTN access route is in place for content providers who do not have an NHS*net* connection, but wish to upload and offer content on NHS*net*,

- service management access exists from the private BT network to the server to enable service and customer operation and management.

Although the three service levels of standard, enhanced and custom are run on a single server, the system, however, is designed so that in the event of declining performance, the service levels can be split across the two, or in the case of failure, the other server can accommodate all three as a hot standby. The servers are both connected to a common storage array with multiple disks for mirroring which holds database and content. Figure 10 gives an overview of the logical design for HealthNet WebWorld.

In support of high availability, redundancy is inherent in the system. Each of the computing platforms has a standby that can be used in the event of failure. In the case of the HealthNet firewalls and the main servers, there are hot standbys that will automatically take over the running of the service when a fault is detected.

In order to comply with NHS security policies [6, 7] and to protect the network, there is a physical separation between the different access networks. This is the reason for the multiple Ethernet hubs, each of which is connected to a different port on the main servers. By this mechanism, there is physical separation of the traffic for HealthNet users, Internet users, content providers and BT management.

The connection between the server infrastructure and HealthNet is a standard direct access connection and is managed as if the service were an NHS organisation. For security classification purposes, BT is classed as a non-NHS third party application provider to the network (see Table 1).

The firewall that connects the local HealthNet router and the internal HealthNet hub routes application level protocols according to the policies in Table 3.

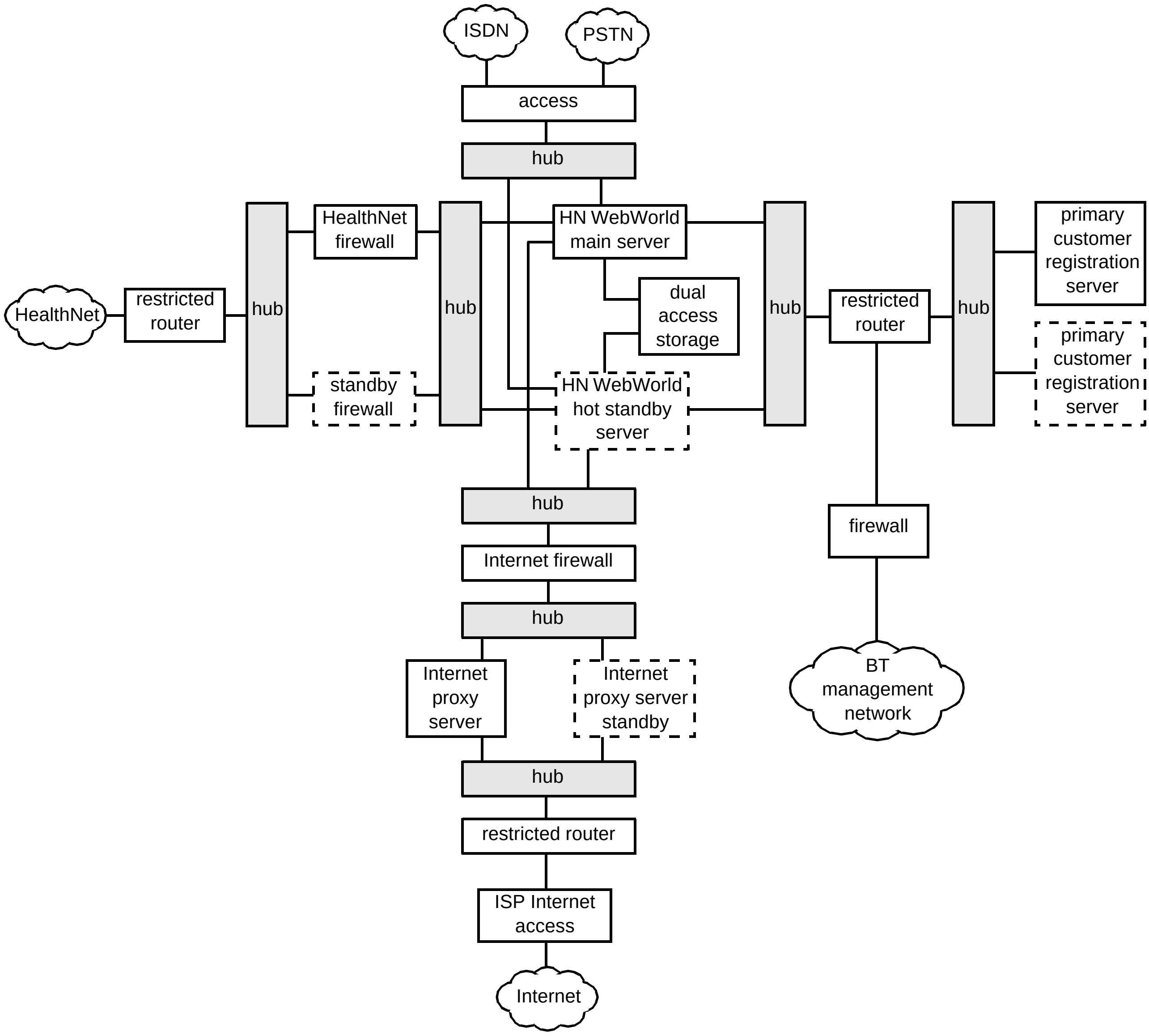

Fig 10 Server platform logical design.

Table 3 Protocols that are allowed to and from the server.

HealthNet to the Server:	FTP, HTTP, SSL, HTTPS
Server to HealthNet:	None permitted (acknowledgements allowed)

Internet content sharing

A proxy server sits outside the firewall as a stand-in for the HealthNet WebWorld main server. Internet clients use a non-HealthNet IP address (and domain name) to access the proxy [13]. The proxy then performs the request on their behalf to the main content servers. The real content resides on the main content server, safely inside the firewall. The proxy server resides outside the firewall, and appears to the Internet client to be the content server. This is called a reverse proxy because proxies are usually used to limit access out from a network, so that users are unable to get to unsavoury sites as prescribed by the system administrator.

This is a solution that is used in BT CampusWorld [14] to restrict access to public sites that are deemed unsuitable for minors. In the HealthNet case, the proxy is used the other way round to restrict access into the network, hence a reverse proxy.

When an Internet client makes a request of the proxy, the proxy server sends the client's request through a specific passage in the firewall to the main server. The main server passes the result through the passage back to the proxy. The proxy sends the retrieved information to the client, as if the proxy were the actual main content server. If the main server returns an error message, the proxy server can intercept the message and change any URLs listed in the headers before sending the message to the client. This prevents external clients from getting redirection URLs to the main server.

The proxy is protected from the Internet by a restricted router, and the main servers are protected from the proxy and the Internet by another firewall. This firewall has a similar configuration as the HealthNet WebWorld firewall and additionally does not allow the upload of information content from the Internet.

Non-NHS information providers connection

As non-NHS information content providers are generally not allowed to connect to NHS *net* as end users [7], they therefore require a secure dedicated access to the server in order to upload content to their Web sites. Access via the Internet is not considered secure enough. Dial-up access is provided over ISDN and PSTN through a dedicated and restricted router and provides a connection directly to the content provider's home area only (FTP and HTTP) (Fig 11 shows this arrangement).

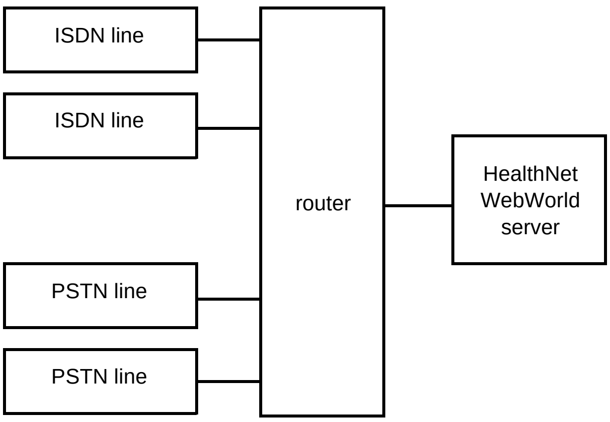

Fig 11 Non-NHS information provider connection.

Security is provided by use of a CHAP authentication server running across both main servers.

Network and service management

The local management network is connected via another firewall to the BT internal network to enable network and service management; access is provided to all possible components of the system and is achieved either by direct Ethernet connection or via a terminal concentrator to serial ports. Service management is performed remotely by the BT on-line customer service centre over the BT internal network so that only authorised users are able to access the service.

Software high-level design

The server software comprises a number of components to support different application types. Based on Sun Solaris, the principle components are those of webserver and database management systems. Figure 12 gives an illustration of the high-level software architecture.

Webservers

The webserver is used to facilitate access to the managed content for the HealthNet WebWorld service. The server supports standard HTTP and secure sockets layer (SSL)-based extensions (HTTPS). This allows for information to be collected from the user in a secure way if required (determined by customer Web page definitions). The server also supports username and password protected access, made available through the enhanced level of service.

A separate webserver is dedicated for Internet access via the proxy with access to a sub-directory (i.e. /internet) of the full customer account. This configuration protects sensitive information from being inadvertently available on the Internet. Each customer account has a set structure depending upon the type of customer that they are. Custom customers will have an extra CGI-bin directory to hold the CGIs to access the database. If the customer has the 'content-sharing' feature enabled the split between HealthNet and Internet content is achieved by providing a sub-directory to the account that is the document root for the Internet webserver. The content is classified as either 'Internet' (/internet) or 'HealthNet' (/healthnet) by means of its location in the user's account.

Figure 13 shows how this is achieved from a process architecture point of view.

5. Conclusion

The cost of delivering healthcare is substantial and the NHS are continuously striving to reduce these costs. The intranet is a new opportunity for reducing costs through investment in information technology and HealthNet is BT's response to delivering the platform on which that investment can take place.

Details of the implementation of the network and a Web-hosting facility have been discussed in this paper, illustrating how typical intranet foundations can be laid. Now that these building blocks are in place, real cost savings and efficiencies can be made through the implementation of additional intranet services [15], applications, information and the re-engineering of business processes to exploit the technology.

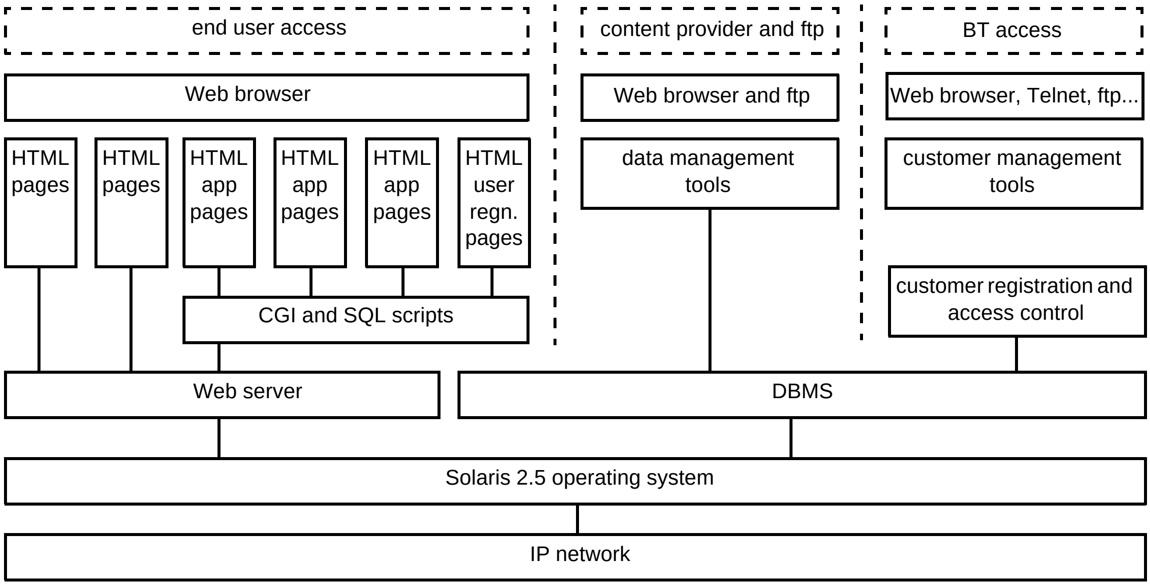

Fig 12 High-level software architecture.

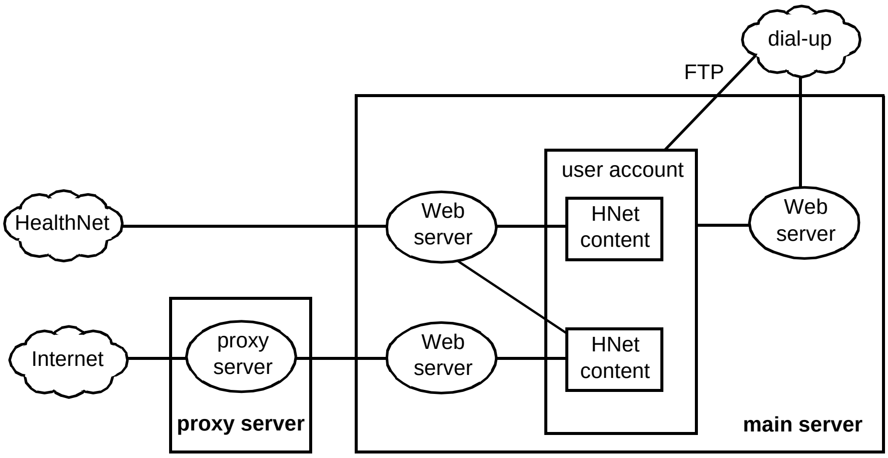

Fig 13 Webserver process architecture.

Acknowledgements

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